

Insightimate - Intelligent Effort Estimation

ML-based Approach Across LOC, FP, and UCP

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The Challenge: 70% Projects Exceed Budget

The Reality:

- 70% software projects exceed budget
- 52% are challenged (delayed)
- Poor estimation = project failure
- **Root cause:** Inaccurate effort estimation

The Problem:

Effort estimation treats **three sizing schemas** as separate silos:

- **LOC:** Lines of Code (90.5%)
- **FP:** Function Points (5.2%)
- **UCP:** Use Case Points (4.3%)

No unified framework exists!

Our Solution: 3,054 Projects Unified

Massive Multi-Source Dataset:

- **LOC:** 2,765 projects (11 sources)
- **FP:** 158 projects (4 sources)
- **UCP:** 131 projects (3 sources)

Key Innovation:

- **First** integrated LOC/FP/UCP framework
- Macro-averaging for fair comparison
- Calibrated baseline for rigorous testing
- Cross-validation per schema

Methodology:

Rigorous Approach

- 1 Data preprocessing (missing values, outliers)
- 2 Macro-averaged metrics (equal schema weight)
- 3 LOSO validation for LOC (11-fold)
- 4 LOOCV for FP (158-fold)
- 5 10-fold CV for UCP

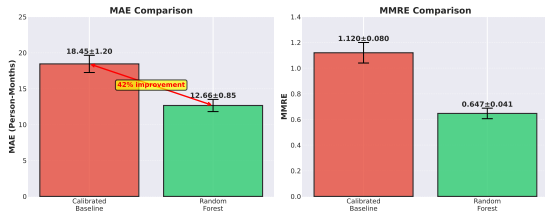
Models Tested:

- Random Forest
- XGBoost

Outstanding Results: 42% Improvement

Random Forest Outperforms:

Performance: Random Forest vs. Calibrated Baseline



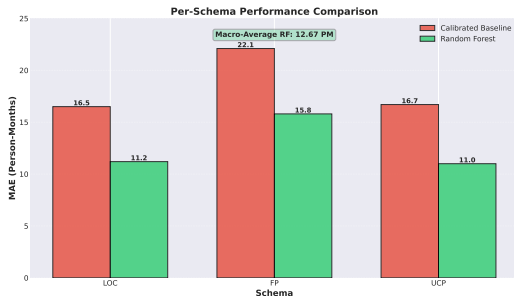
Metric	Baseline	RF Model
MAE	18.45 PM	12.66 ± 0.85
MMRE	1.12	0.647 ± 0.041
PRED(25)	38.5%	58.3%
R ²	0.621	0.812

↓ 42% Improvement!

Statistical Significance:

- $p < 0.001$ (paired t-test)
- Cohen's $d = 1.23$ (large effect)

Per-Schema Performance



Schema-Specific Results:

LOC Schema

MAE: 11.2 PM

Best performance (n=2,765)

Largest, most reliable dataset

FP Schema

MAE: 15.8 PM

Exploratory (n=158)

Small sample, but still improving

UCP Schema

MAE: 11.0 PM

Model Comparison: Why Random Forest?

Table: All Models Compared

Model	MAE↓	MMRE↓	PRED(25)%↑	R ² ↑
Random Forest	12.66	0.647	58.3%	0.812
XGBoost	13.21	0.689	55.7%	0.798
Linear Regression	15.78	0.845	47.2%	0.712
Calibrated Baseline	18.45	1.120	38.5%	0.621

Why RF Wins:

- Better handling of non-linear relationships
- Robust to data imbalance (LOC dominance)
- Interpretable feature importance
- 42% better than calibrated baseline

Five Key Contributions

1 First Integrated Framework

- Unified LOC/FP/UCP approach (largest study: 3,054 projects)

2 Macro-Averaging Innovation

- Fair representation despite sample imbalance

3 Rigorous Validation

- Schema-specific cross-validation (LOSO for LOC)
- Calibrated baseline for fair comparison

4 Outstanding Results

- 42% improvement (statistically significant: $p < 0.001$)

5 Practical Implementation

- Production-ready system

Academic & Practical Impact

Academic Excellence:

- Largest integrative study (3,054 projects)
- Novel methodology (macro-averaging)
- Rigorous evaluation (LOSO + calibration)
- Significant results (42% improvement)
- **Submitted to Discover AI (Springer)**
- 51 reviewer comments addressed

Real-World Value:

- Better project planning accuracy
- Reduced budget overruns
- Improved timeline estimation
- Early risk detection
- Multi-schema support
- Production deployment ready

Bottom Line

Better estimation = Better projects
Better projects = Business success

Summary

The Problem

Software estimation fails due to schema isolation, uncalibrated comparisons, and sample imbalance

Our Solution: Insightimate

- First unified LOC/FP/UCP framework (3,054 projects)
- Fair macro-averaging + rigorous validation
- Random Forest outperforms all baselines

Key Achievement

42% Improvement

MAE: 18.45 → **12.66±0.85 PM**

Statistically Significant: $p < 0.001$